LinkWise: A RAG-Based Digital Safety Guidance Chatbot for Assessing Unfamiliar URLs Among Students in Secondary and Higher Education

Balneg, Maria Cristina B.
Department of Computer Studies and Engineering
Jose Rizal University
Mandaluyong City, Philippines
mariacristina.balneg@my.jru.edu

Sayat, Lyona Yvonne A.
Department of Computer Studies and Engineering
Jose Rizal University
Mandaluyong City, Philippines
lyonneyvonne.sayat@my.jru.edu

Bartolome, Beatrice Matilda D.
Department of Computer Studies and Engineering
Jose Rizal University
Mandaluyong City, Philippines
beatricematilda.bartolome@my.jru.edu

***Abstract*—This concept paper proposes LinkWise, a Retrieval-Augmented Generation (RAG) chatbot that gives students digital-safety guidance when they encounter unfamiliar URLs. The study is linked to Sustainable Development Goal 4 because safe and responsible use of digital technology is part of practical digital literacy. The chatbot will retrieve information from trusted cybersecurity and privacy documents before generating a response. The proposed study will compare RAG-based answers with answers from a chatbot without RAG using 20-30 controlled test questions. Responses will be checked for correctness, groundedness, completeness, and proper handling of unsupported questions. The project focuses on guidance and education, not automatic malicious-URL detection.**

Keywords—digital safety, phishing, RAG, suspicious URLs, SDG 4, student cybersecurity

I. SELECTED UN SGD AND JUSTIFICTION

A. Selected SGD and Relevant

The proposed study is connected to Sustainable Development Goal 4: Quality Education. It is especially related to SDG Target 4.4, which includes information and communications technology (ICT) skills as an important skill area for young people and adults [1]. For this study, digital literacy includes knowing how to use online platforms safely and responsibly.

B. Justification

Students in secondary and higher education use messaging apps, social media, email, and online learning platforms for school and communication. While using these platforms, they may receive unfamiliar URLs. Some links are legitimate, while others may lead to phishing pages, fake websites, or other harmful content. Students may not always know what warning signs to check, how to assess a link without opening it, or what action to take when a link looks suspicious. This problem is relevant to SDG 4 because students need practical digital-safety skills as part of responsible technology use.

C. Connection to RAG

A RAG chatbot can provide guidance that is based on trusted documents instead of relying only on the language model's built-in knowledge. When a student asks about an unfamiliar URL or suspicious message, the system can retrieve relevant cybersecurity guidance and use it as the basis of the answer. The chatbot can explain common warning signs, safe ways to verify a message or website, risks of shortened URLs, and steps to take after clicking a suspicious link. It will not guarantee that a live URL is safe or malicious and will not replace specialized URL-scanning or malware-detection tools.

II. NARROW PROBLEM STATEMENT

The U.S. Federal Trade Commission (FTC) guide "How To Recognize and Avoid Phishing Scams" gives examples of phishing messages and practical advice for handling suspicious links and requests for personal information [4]. It is a public government resource and can provide additional guidance for the knowledge base.

These are initial sources only. The group may add other trusted and publicly available cybersecurity documents after checking their relevance, date, accessibility, and permission for use.

Broad domain: Digital safety and digital literacy.
SDG-related issue: Safe and responsible use of online information and communication technologies.
Target users: Students in secondary and higher education.
Narrow problem: Students may have difficulty recognizing warning signs and deciding what safety action to take when they encounter unfamiliar URLs through messaging apps, social media, email, online platforms, and other digital environments.
Proposed chatbot focus: Provide document-grounded guidance on suspicious URL and message characteristics, safe checking practices, phishing avoidance, and appropriate actions when a link appears suspicious.

Students may face online risks because it is not always easy to judge an unfamiliar link before opening it. The proposed study will investigate whether a RAG-based chatbot can make trusted digital-safety information easier to access and understand. The study is limited to guidance about unfamiliar or suspicious URLs and does not attempt to solve cybersecurity as a whole.

III. PRIMARY RESEARCH QUESTION

The primary research question is: To what extent can a RAG-based chatbot provide more correct and well-supported digital-safety guidance for students in secondary and higher education who encounter unfamiliar URLs compared with a chatbot without RAG?

This question is focused and measurable because the same set of questions can be given to two systems: a RAG-

based chatbot that retrieves information from the knowledge base, and a non-RAG chatbot that answers without retrieval. Their responses can be compared based on correctness, groundedness, completeness, and proper handling of questions that are not supported by the knowledge base.

IV. Initial Knowledge-Base Data Sources

A. National Privacy Commission: Smishing Guidance

The National Privacy Commission (NPC) bulletin "Preventive Data Privacy Practices Against Smishing" is an official Philippine government source. It explains smishing, suspicious and shortened links, fraudulent sites, risks to personal information, and recommended safety practices [2]. It is publicly available and can provide local guidance for the chatbot.

B. National Privacy Commission: Security and Privacy Threaths

The NPC webpage "Threats to Security and Privacy" explains phishing, pharming, malware, fake websites, and other online risks [3]. It is publicly accessible and can support answers about common digital threats and safe online behavior.

C. Federal Trade Commison: Phissing Guidance

The problem is narrow and manageable because the study does not try to detect every malicious website or solve all cybersecurity problems. It focuses on one information need: helping students understand what to check and what to do when they encounter an unfamiliar or suspicious URL. The knowledge base can be limited to a small set of trusted cybersecurity and privacy documents, while the evaluation can use 20-30 safe test scenarios. This scope allows the group to study whether RAG improves the quality of digital-safety guidance without accessing real malicious websites or building a full malicious-URL detection system.

V. Initial Evaluation Data Plan

The group plans to prepare about 20-30 safe and controlled test questions or scenarios. These questions will represent situations that students may experience through messaging apps, social media, email, and online learning platforms. The test set may include shortened URLs, misspelled domains, unexpected messages, requests for passwords or personal information, urgent messages, links that appear to come from schools or banks, and questions about what to do after clicking a suspicious link.

A sample scenario is a message saying that a package could not be delivered and asking the student to click a shortened URL immediately. A possible test question is: "What should the student do before opening the link?" The expected answer and supporting passage will be based on the selected knowledge-base documents.

A small number of questions will also be intentionally outside the knowledge base. These items will test whether the chatbot can admit that there is not enough information instead of making up an unsupported answer. The group will not require students or researchers to visit real malicious websites. The evaluation will compare RAG and non-RAG responses using correctness, groundedness, completeness, and handling of unsupported questions.

VI. Possible Language Model and Justifaction

A. Proposed Model

The proposed language model is GPT-OSS 20B (openai/gpt-oss-20b) through Groq. Groq lists the model as a production model with text input and output and a context window of 131,072 tokens [5]. This makes it a practical option for a student RAG experiment that needs to process retrieved passages together with a user question.

B. Expected Role

The model will receive the student's question and the relevant passages retrieved from the knowledge base. It will then turn the retrieved information into a clear and understandable answer. It should explain the recommended safety action and state when the available documents do not contain enough information to answer confidently.

C. Practical Considerations

The first version of the study may focus on English questions and English cybersecurity documents. Filipino or Taglish questions can be added later if the group decides to test them separately. GPT-OSS 20B is only a preliminary model choice for the proposal and should not be described as the best model without comparative evidence.

VII. Group Reflection

The problem is narrow and manageable because the study does not try to detect every malicious website or solve all cybersecurity problems. It focuses on one information need: helping students understand what to check and what to do when they encounter an unfamiliar or suspicious URL. The knowledge base can be limited to a small set of trusted cybersecurity and privacy documents, while the evaluation can use 20-30 safe test scenarios. This scope allows the group to study whether RAG improves the quality of digital-safety guidance without accessing real malicious websites or building a full malicious-URL detection system.

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